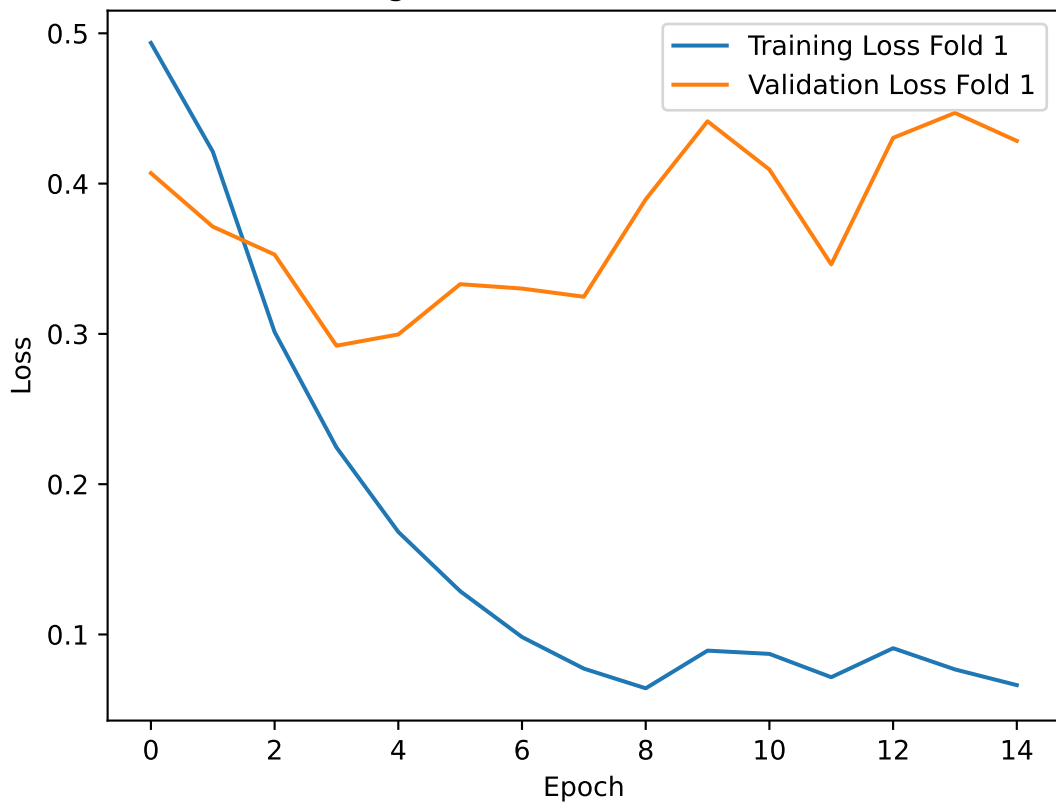


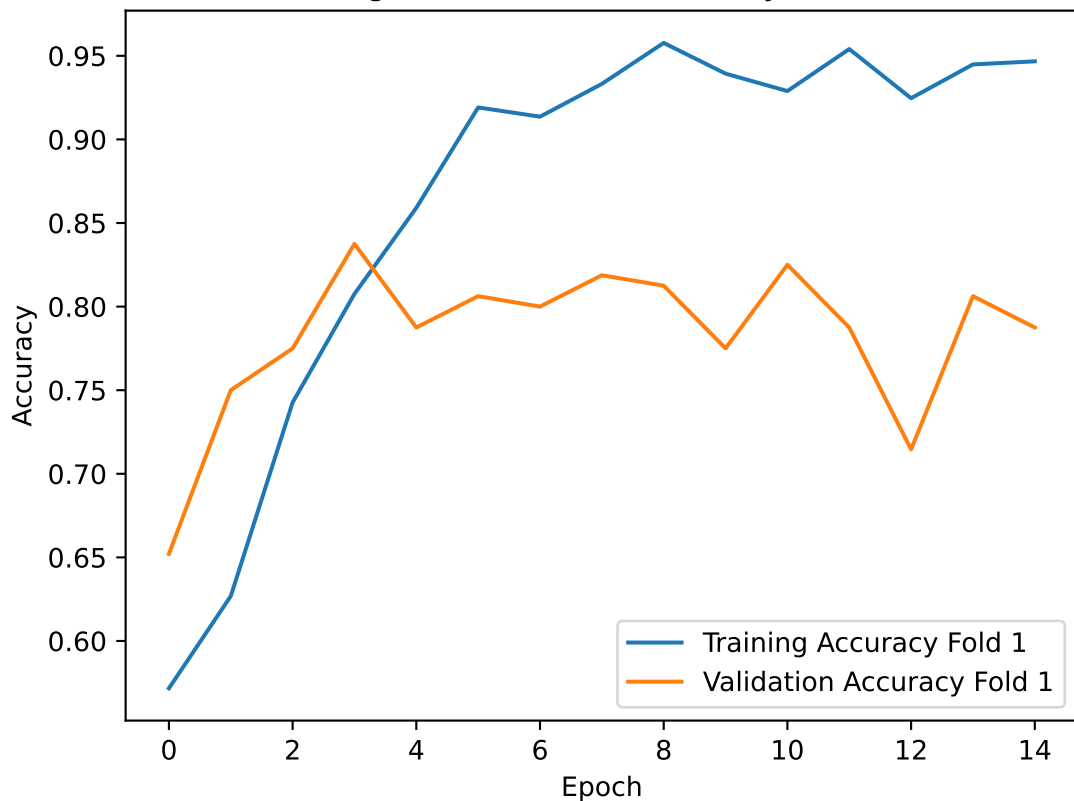
Validation F1-Score for fold 1: 0.06716417910447761

0.06716417910447761

Training and Validation Loss for Fold 1



Training and Validation Accuracy for Fold 1



Best Hyperparameters for Fold 1:

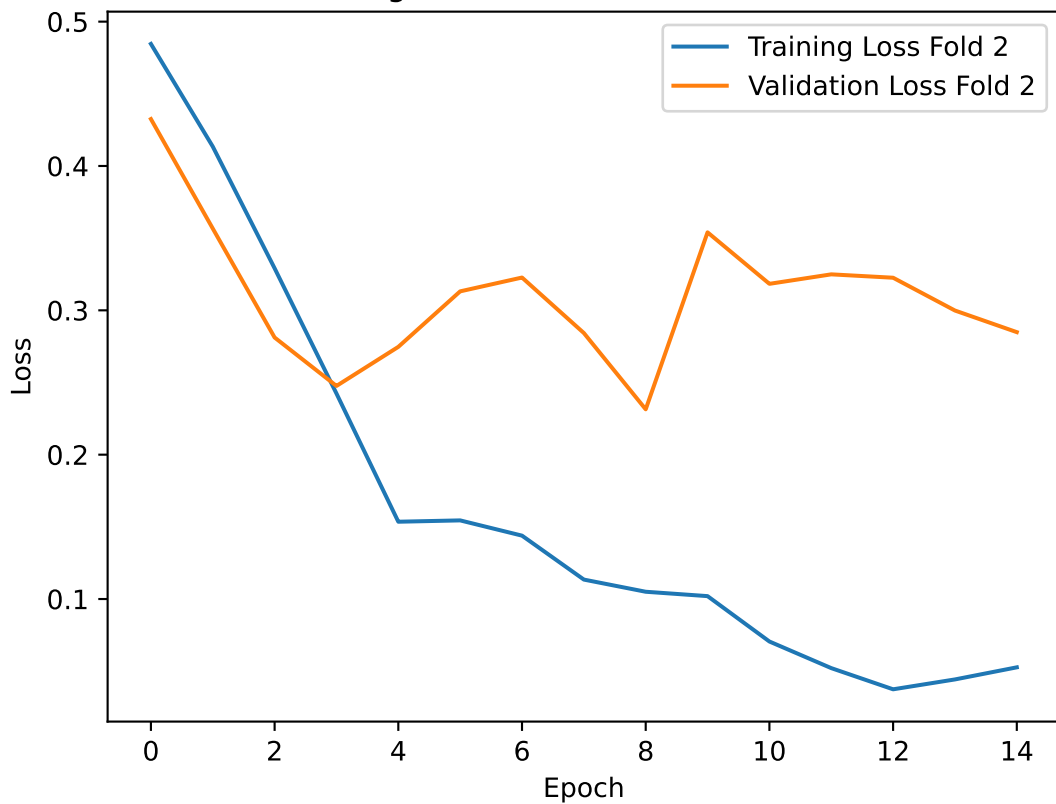
```
{'conv_filters': 32, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}
```

Best Trial Score for Fold 1: 0.84375

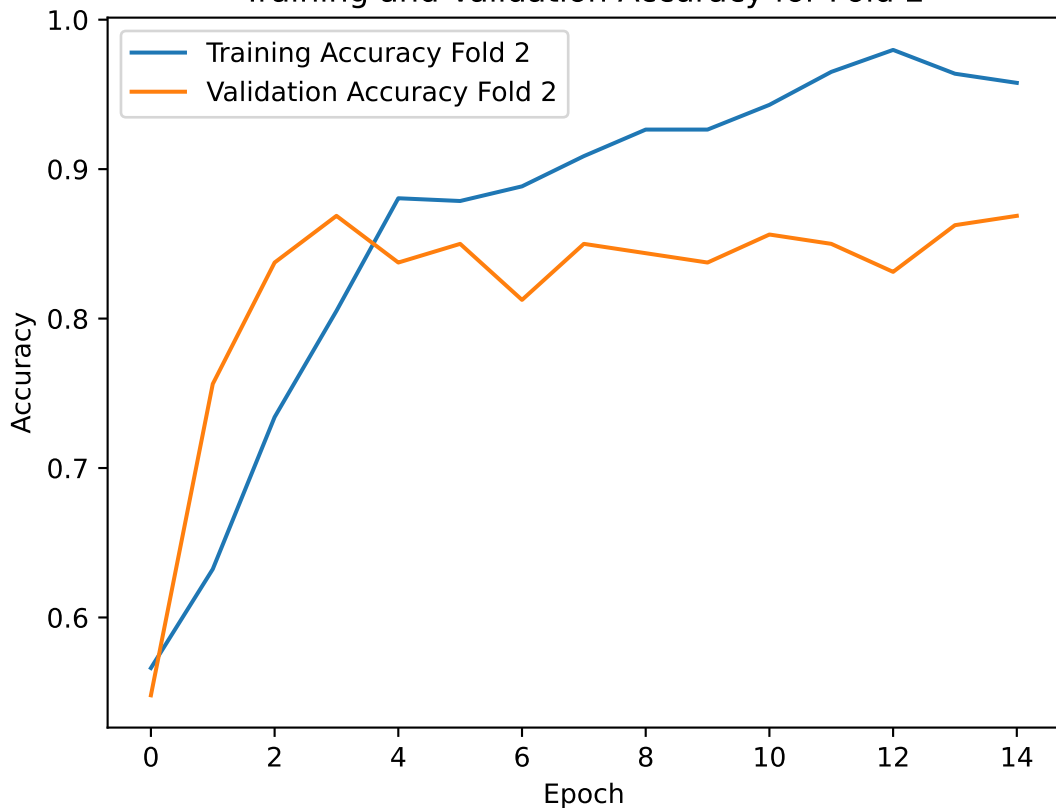
Validation F1-Score for fold 2: 0.08823529411764705

0.08823529411764705

Training and Validation Loss for Fold 2



Training and Validation Accuracy for Fold 2



Best Hyperparameters for Fold 2:

```
{'conv_filters': 192, 'conv_activation': 'relu', 'unitsD1': 416, 'unitsD2': 32}
```

Best Trial Score for Fold 2: 0.893750011920929

Search: Running Trial #1

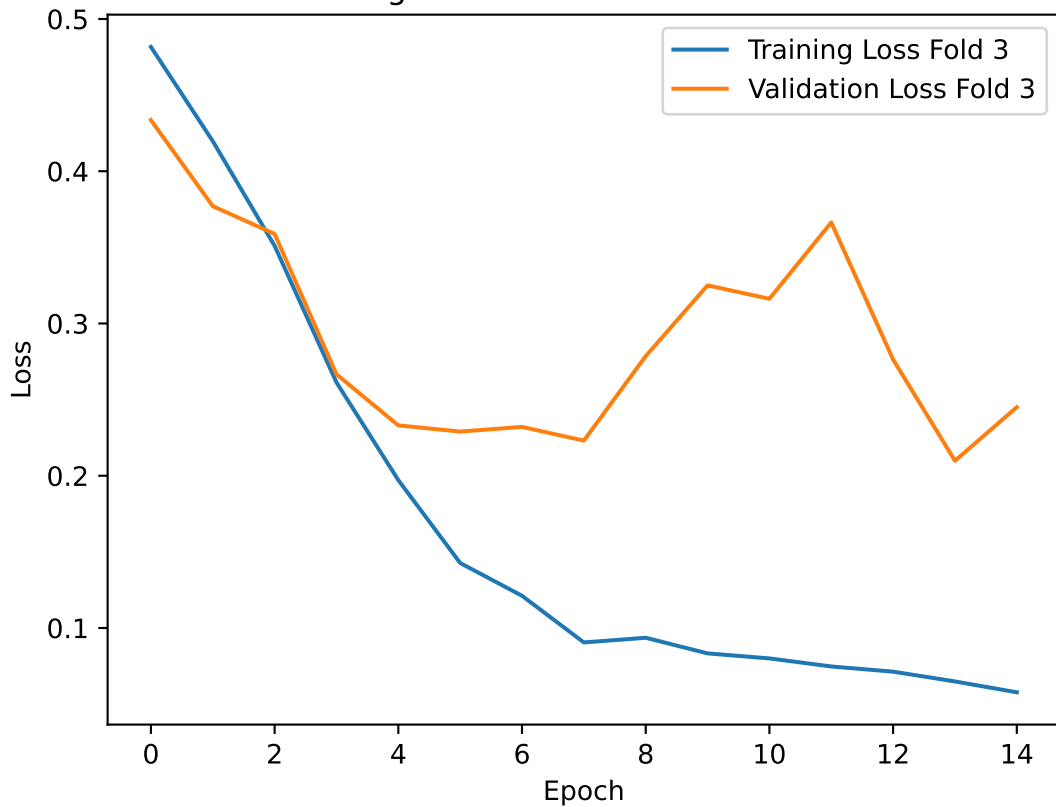
Value	Best Value So Far	Hyperparameter
96	96	conv_filters
relu	relu	conv_activation
160	160	unitsD1
32	32	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m[37m-----[0m [1m10:07][0m 38s/step - accuracy: 0.5000 - loss: 0.4898
[32m][0m[37m-----[0m [1m3s][0m 257ms/step - accuracy: 0.4766 - loss: 0.4988
[32m][0m[37m-----[0m [1m3s][0m 255ms/step - accuracy: 0.4740 - loss: 0.4966
[32m][0m[37m-----[0m [1m3s][0m 255ms/step - accuracy: 0.4824 - loss: 0.4954
[32m][0m[37m-----[0m [1m3s][0m 255ms/step - accuracy: 0.4947 - loss: 0.4933
[32m][0m[37m-----[0m [1m2s][0m 254ms/step - accuracy: 0.4973 - loss: 0.4967
[32m][0m[37m-----[0m [1m2s][0m 254ms/step - accuracy: 0.4996 - loss: 0.4982
[32m][0m[37m-----[0m [1m2s][0m 254ms/step - accuracy: 0.5084 - loss: 0.4923
[32m][0m[37m-----[0m [1m2s][0m 254ms/step - accuracy: 0.5152 - loss: 0.4878
[32m][0m[37m-----[0m [1m1s][0m 254ms/step - accuracy: 0.5196 - loss: 0.4845
[32m][0m[37m-----[0m [1m1s][0m 254ms/step - accuracy: 0.5235 - loss: 0.4821
[32m][0m[37m-----[0m [1m1s][0m 253ms/step - accuracy: 0.5261 - loss: 0.4805
[32m][0m[37m-----[0m [1m1s][0m 254ms/step - accuracy: 0.5278 - loss: 0.4794
[32m][0m[37m-----[0m [1m0s][0m 254ms/step - accuracy: 0.5295 - loss: 0.4785
[32m][0m[37m-----[0m [1m0s][0m 254ms/step - accuracy: 0.5309 - loss: 0.4779
[32m][0m[37m-----[0m [1m0s][0m 254ms/step - accuracy: 0.5316 - loss: 0.4778
[32m][0m[37m[0m [1m0s][0m 826ms/step - accuracy: 0.5330 - loss: 0.4775
[32m][0m[37m[0m [1m71s][0m 2s/step - accuracy: 0.5342 - loss: 0.4773 - val_accuracy: 0.6208 - val_loss: 0.5590
Epoch 2/15
[1m 1/17][0m [32m][0m[37m-----[0m [1m4s][0m 312ms/step - accuracy: 0.5000 - loss: 0.5952
[32m][0m[37m-----[0m [1m3s][0m 253ms/step - accuracy: 0.4844 - loss: 0.6433
[32m][0m[37m-----[0m [1m3s][0m 253ms/step - accuracy: 0.4826 - loss: 0.6447
[32m][0m[37m-----[0m [1m3s][0m 253ms/step - accuracy: 0.4831 - loss: 0.6336
[32m][0m[37m-----[0m [1m3s][0m 254ms/step - accuracy: 0.4865 - loss: 0.6208
[32m][0m[37m-----[0m [1m2s][0m 255ms/step - accuracy: 0.4844 - loss: 0.6272
[32m][0m[37m-----[0m [1m2s][0m 255ms/step - accuracy: 0.4879 - loss: 0.6235
[32m][0m[37m-----[0m [1m2s][0m 255ms/step - accuracy: 0.4972 - loss: 0.6124
[32m][0m[37m-----[0m [1m2s][0m 255ms/step - accuracy: 0.5076 - loss: 0.6015
[32m][0m[37m-----[0m [1m1s][0m 255ms/step - accuracy: 0.5162 - loss: 0.5924
[32m][0m[37m-----[0m [1m1s][0m 255ms/step - accuracy: 0.5230 - loss: 0.5852
[32m][0m[37m-----[0m [1m1s][0m 256ms/step - accuracy: 0.5273 - loss: 0.5794
[32m][0m[37m-----[0m [1m1s][0m 256ms/step - accuracy: 0.5304 - loss: 0.5742
[32m][0m[37m-----[0m [1m0s][0m 256ms/step - accuracy: 0.5333 - loss: 0.5691
[32m][0m[37m-----[0m [1m0s][0m 256ms/step - accuracy: 0.5370 - loss: 0.5642
[32m][0m[37m-----[0m [1m0s][0m 256ms/step - accuracy: 0.5405 - loss: 0.5596
[32m][0m[37m[0m [1m0s][0m 247ms/step - accuracy: 0.5439 - loss: 0.5552
[32m][0m[37m[0m [1m6s][0m 370ms/step - accuracy: 0.5469 - loss: 0.5513 - val_accuracy: 0.7125 - val_loss: 0.4229
Epoch 3/15

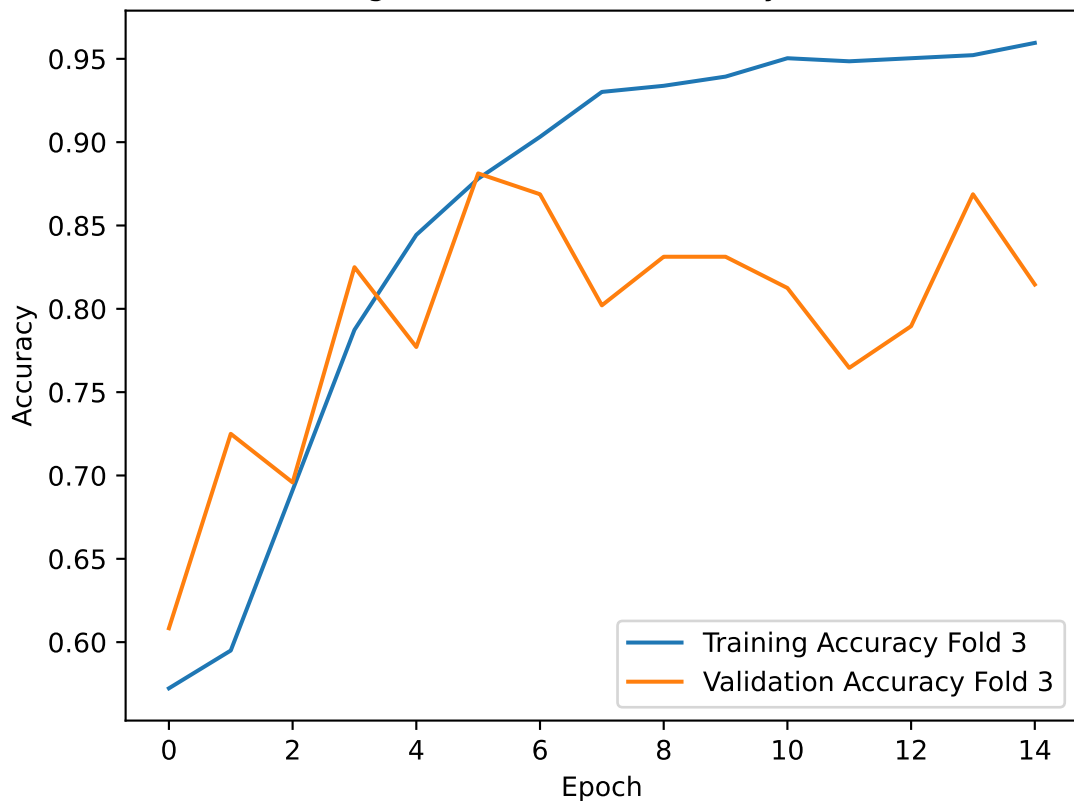
Validation F1-Score for fold 3: 0.15714285714285714

0.15714285714285714

Training and Validation Loss for Fold 3



Training and Validation Accuracy for Fold 3



Best Hyperparameters for Fold 3:

```
{'conv_filters': 128, 'conv_activation': 'relu', 'unitsD1': 160, 'unitsD2': 32}
```

Best Trial Score for Fold 3: 0.8812500238418579

Search: Running Trial #1

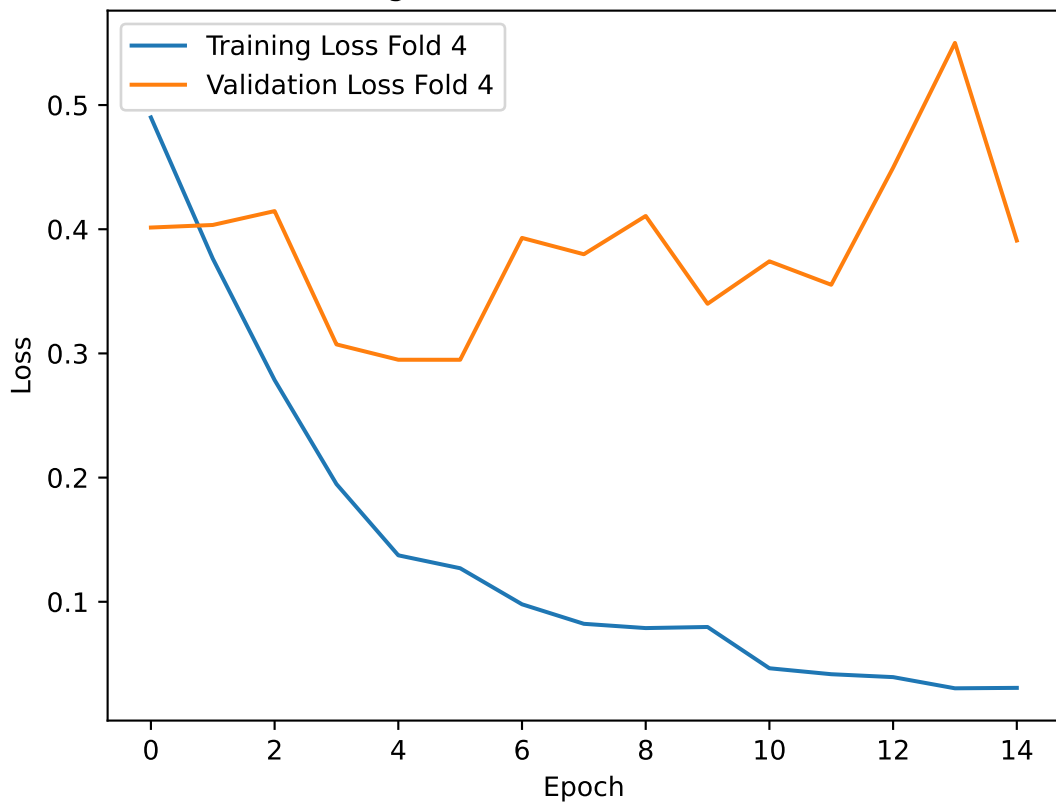
Value	Best Value So Far	Hyperparameter
128	128	conv_filters
relu	relu	conv_activation
160	160	unitsD1
32	32	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m][37m-----][0m [1m10:07][0m 38s/step - accuracy: 0.4375 - loss: 0.5188
[32m][0m][37m-----][0m [1m3s][0m 257ms/step - accuracy: 0.4453 - loss: 0.5234
[32m][0m][37m-----][0m [1m3s][0m 255ms/step - accuracy: 0.4358 - loss: 0.5284
[32m][0m][37m-----][0m [1m3s][0m 254ms/step - accuracy: 0.4362 - loss: 0.5273
[32m][0m][37m-----][0m [1m3s][0m 253ms/step - accuracy: 0.4465 - loss: 0.5241
[32m][0m][37m-----][0m [1m2s][0m 252ms/step - accuracy: 0.4510 - loss: 0.5244
[32m][0m][37m-----][0m [1m2s][0m 252ms/step - accuracy: 0.4580 - loss: 0.5225
[32m][0m][37m-----][0m [1m2s][0m 252ms/step - accuracy: 0.4711 - loss: 0.5132
[32m][0m][37m-----][0m [1m2s][0m 252ms/step - accuracy: 0.4797 - loss: 0.5063
[32m][0m][37m-----][0m [1m1s][0m 252ms/step - accuracy: 0.4855 - loss: 0.5017
[32m][0m][37m-----][0m [1m1s][0m 252ms/step - accuracy: 0.4902 - loss: 0.4982
[32m][0m][37m-----][0m [1m1s][0m 252ms/step - accuracy: 0.4942 - loss: 0.4959
[32m][0m][37m-----][0m [1m1s][0m 253ms/step - accuracy: 0.4969 - loss: 0.4942
[32m][0m][37m-----][0m [1m0s][0m 253ms/step - accuracy: 0.4992 - loss: 0.4928
[32m][0m][37m-----][0m [1m0s][0m 253ms/step - accuracy: 0.5008 - loss: 0.4918
[32m][0m][37m-----][0m [1m0s][0m 253ms/step - accuracy: 0.5026 - loss: 0.4910
[32m][0m][37m][0m [1m0s][0m 758ms/step - accuracy: 0.5049 - loss: 0.4901
[32m][0m][37m][0m [1m69s][0m 2s/step - accuracy: 0.5070 - loss: 0.4893 - val_accuracy: 0.6562 - val_loss: 0.4407
Epoch 2/15
[1m 1/17][0m [32m][0m][37m-----][0m [1m5s][0m 321ms/step - accuracy: 0.4375 - loss: 0.5574
[32m][0m][37m-----][0m [1m3s][0m 256ms/step - accuracy: 0.4688 - loss: 0.5712
[32m][0m][37m-----][0m [1m3s][0m 253ms/step - accuracy: 0.4722 - loss: 0.5714
[32m][0m][37m-----][0m [1m3s][0m 252ms/step - accuracy: 0.4733 - loss: 0.5627
[32m][0m][37m-----][0m [1m3s][0m 252ms/step - accuracy: 0.4786 - loss: 0.5542
[32m][0m][37m-----][0m [1m2s][0m 253ms/step - accuracy: 0.4822 - loss: 0.5477
[32m][0m][37m-----][0m [1m2s][0m 253ms/step - accuracy: 0.4918 - loss: 0.5366
[32m][0m][37m-----][0m [1m2s][0m 253ms/step - accuracy: 0.5055 - loss: 0.5217
[32m][0m][37m-----][0m [1m2s][0m 253ms/step - accuracy: 0.5203 - loss: 0.5076
[32m][0m][37m-----][0m [1m1s][0m 253ms/step - accuracy: 0.5317 - loss: 0.4965
[32m][0m][37m-----][0m [1m1s][0m 253ms/step - accuracy: 0.5412 - loss: 0.4869
[32m][0m][37m-----][0m [1m1s][0m 253ms/step - accuracy: 0.5484 - loss: 0.4792
[32m][0m][37m-----][0m [1m1s][0m 254ms/step - accuracy: 0.5534 - loss: 0.4736
[32m][0m][37m-----][0m [1m0s][0m 254ms/step - accuracy: 0.5574 - loss: 0.4693
[32m][0m][37m-----][0m [1m0s][0m 253ms/step - accuracy: 0.5600 - loss: 0.4659
[32m][0m][37m-----][0m [1m0s][0m 253ms/step - accuracy: 0.5627 - loss: 0.4632
[32m][0m][37m][0m [1m0s][0m 245ms/step - accuracy: 0.5659 - loss: 0.4607
[32m][0m][37m][0m [1m6s][0m 374ms/step - accuracy: 0.5687 - loss: 0.4586 - val_accuracy: 0.6896 - val_loss: 0.3844
Epoch 3/15

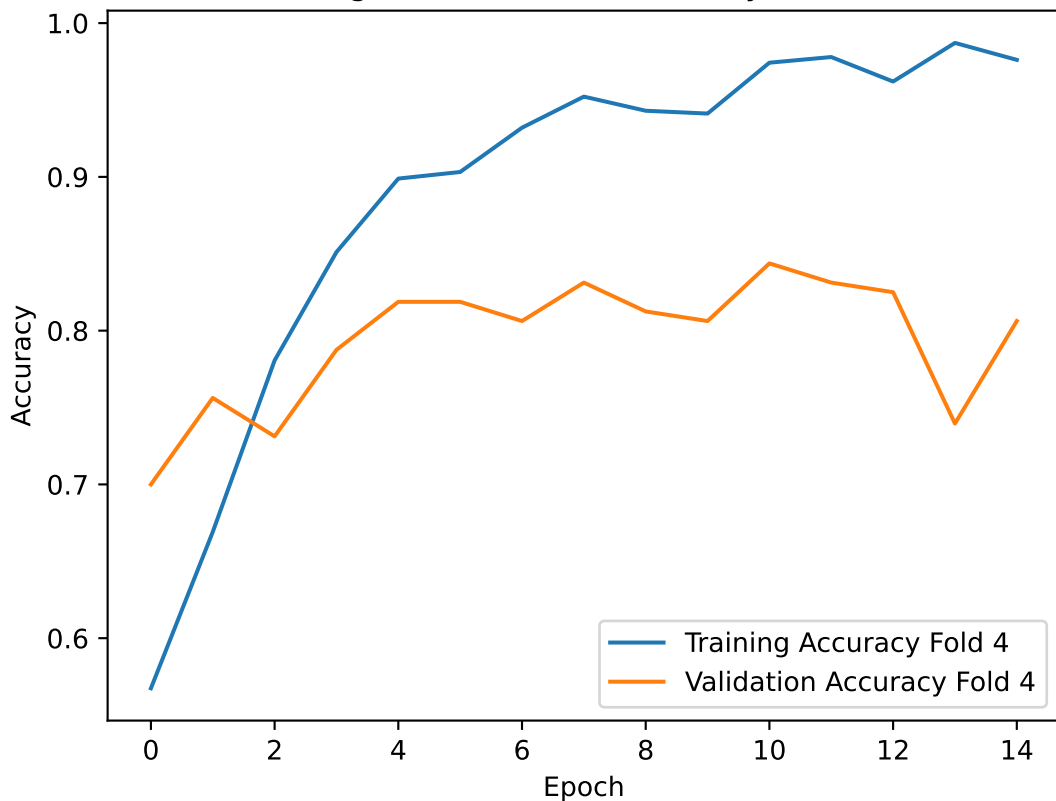
Validation F1-Score for fold 4: 0.045283018867924525

0.045283018867924525

Training and Validation Loss for Fold 4



Training and Validation Accuracy for Fold 4



Best Hyperparameters for Fold 4:

`{'conv_filters': 224, 'conv_activation': 'relu', 'unitsD1': 416, 'unitsD2': 160}`

Best Trial Score for Fold 4: 0.831250011920929

Search: Running Trial #1

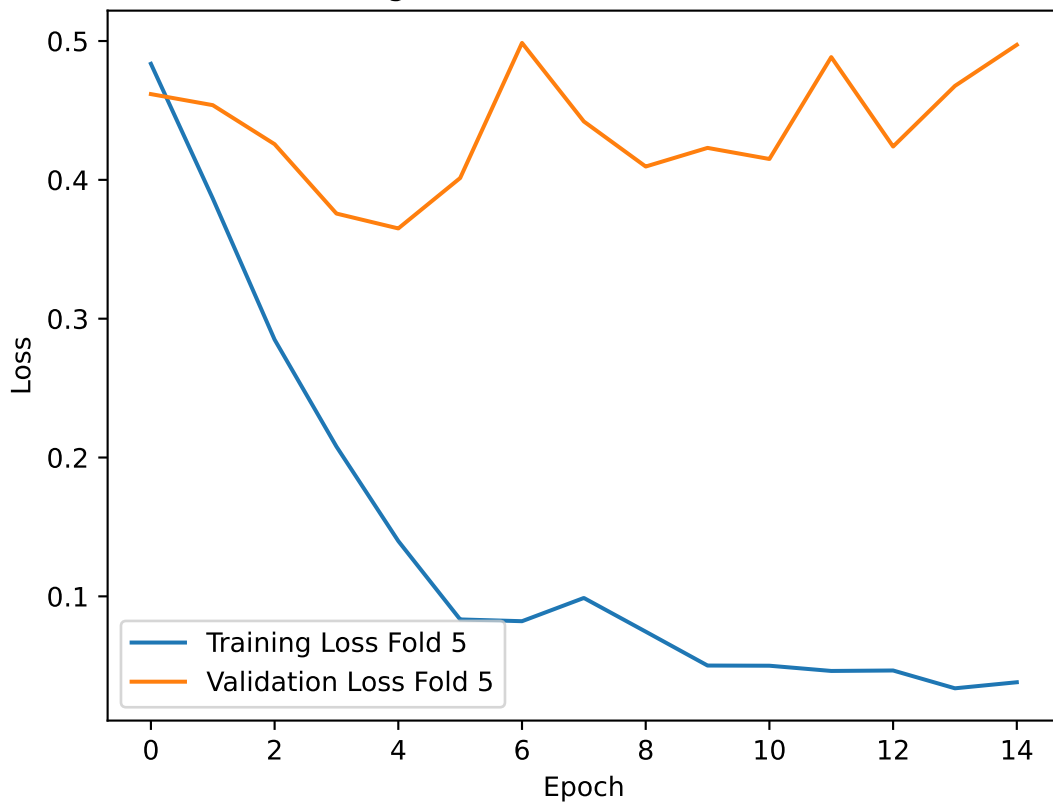
Value	Best Value So Far	Hyperparameter
224	224	conv_filters
relu	relu	conv_activation
416	416	unitsD1
160	160	unitsD2

Epoch 1/15
[0m 1s 17m 0s][32m 0s][0m 0s][37m 0s] [0m 0s][1m 10:27][0m 39s/step - accuracy: 0.5312 - loss: 0.4999
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 4s][0m 268ms/step - accuracy: 0.5391 - loss: 0.5219
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 3s][0m 268ms/step - accuracy: 0.5191 - loss: 0.5370
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 3s][0m 266ms/step - accuracy: 0.5124 - loss: 0.5409
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 3s][0m 266ms/step - accuracy: 0.5199 - loss: 0.5376
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 266ms/step - accuracy: 0.5235 - loss: 0.5367
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 265ms/step - accuracy: 0.5265 - loss: 0.5335
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 265ms/step - accuracy: 0.5340 - loss: 0.5258
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 265ms/step - accuracy: 0.5410 - loss: 0.5184
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 265ms/step - accuracy: 0.5453 - loss: 0.5134
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 265ms/step - accuracy: 0.5477 - loss: 0.5097
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 265ms/step - accuracy: 0.5491 - loss: 0.5066
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 265ms/step - accuracy: 0.5487 - loss: 0.5054
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 265ms/step - accuracy: 0.5487 - loss: 0.5038
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 265ms/step - accuracy: 0.5484 - loss: 0.5025
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 265ms/step - accuracy: 0.5484 - loss: 0.5011
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 849ms/step - accuracy: 0.5488 - loss: 0.4996
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 74s][0m 2s/step - accuracy: 0.5492 - loss: 0.4983 - val_accuracy: 0.6333 - val_loss: 0.4376
Epoch 2/15
[0m 1s 17m 0s][32m 0s][0m 0s][37m 0s] [0m 0s][1m 5s][0m 331ms/step - accuracy: 0.4375 - loss: 0.5700
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 4s][0m 267ms/step - accuracy: 0.4844 - loss: 0.5496
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 3s][0m 268ms/step - accuracy: 0.4896 - loss: 0.5548
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 3s][0m 267ms/step - accuracy: 0.5039 - loss: 0.5467
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 3s][0m 267ms/step - accuracy: 0.5194 - loss: 0.5363
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 266ms/step - accuracy: 0.5300 - loss: 0.5264
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 266ms/step - accuracy: 0.5423 - loss: 0.5146
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 266ms/step - accuracy: 0.5575 - loss: 0.4988
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 2s][0m 266ms/step - accuracy: 0.5728 - loss: 0.4831
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 266ms/step - accuracy: 0.5845 - loss: 0.4709
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 266ms/step - accuracy: 0.5949 - loss: 0.4605
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 266ms/step - accuracy: 0.6031 - loss: 0.4520
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 1s][0m 266ms/step - accuracy: 0.6085 - loss: 0.4458
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 266ms/step - accuracy: 0.6128 - loss: 0.4410
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 266ms/step - accuracy: 0.6168 - loss: 0.4366
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 266ms/step - accuracy: 0.6205 - loss: 0.4329
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 0s][0m 257ms/step - accuracy: 0.6243 - loss: 0.4293
[32m 0s][0m 0s][37m 0s] [0m 0s][1m 5s][0m 311ms/step - accuracy: 0.6277 - loss: 0.4261 - val_accuracy: 0.6333 - val_loss: 0.4208
Epoch 3/15

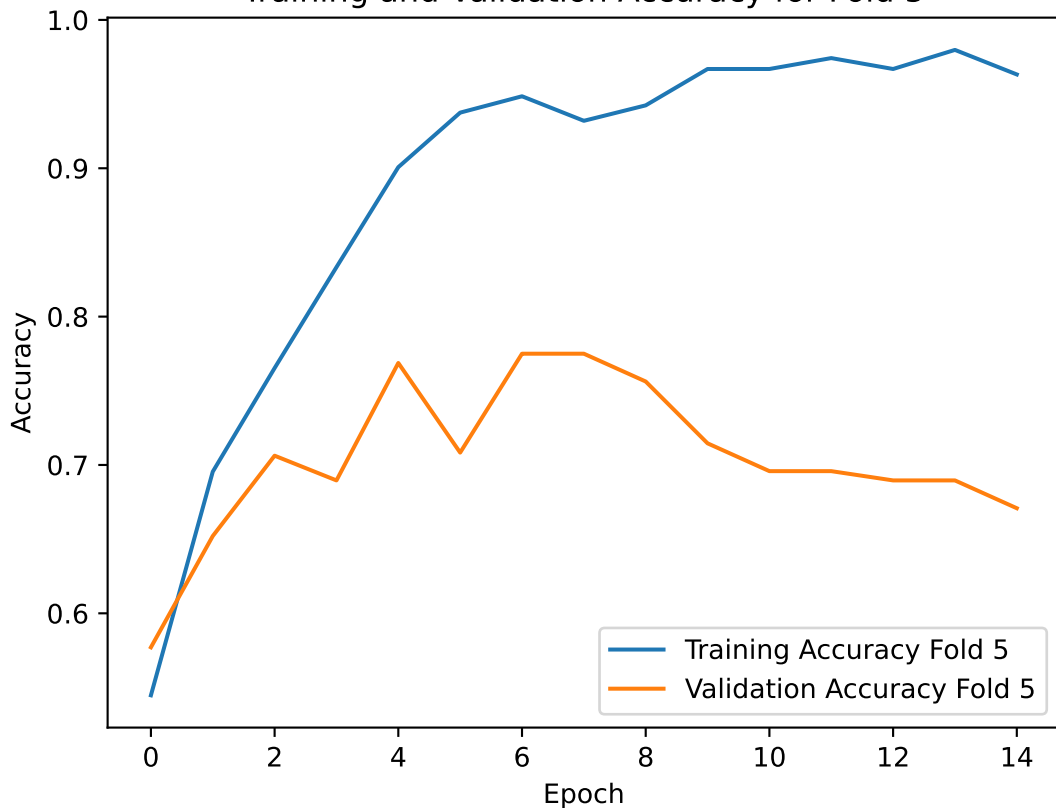
Validation F1-Score for fold 5: 0.125

0.125

Training and Validation Loss for Fold 5



Training and Validation Accuracy for Fold 5



Best Hyperparameters for Fold 5:

```
{'conv_filters': 128, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}
```

Best Trial Score for Fold 5: 0.8500000238418579

Search: Running Trial #1

Value	Best Value So Far	Hyperparameter
128	128	conv_filters
relu	relu	conv_activation
288	288	unitsD1
32	32	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m][37m-----][0m [1m10:13][0m 38s/step - accuracy: 0.4375 - loss: 0.5197
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.4375 - loss: 0.5209
[32m][0m][37m-----][0m [1m3s][0m 260ms/step - accuracy: 0.4444 - loss: 0.5149
[32m][0m][37m-----][0m [1m3s][0m 261ms/step - accuracy: 0.4564 - loss: 0.5101
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.4714 - loss: 0.5057
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.4753 - loss: 0.5064
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.4782 - loss: 0.5071
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.4882 - loss: 0.5006
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.4972 - loss: 0.4957
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5028 - loss: 0.4926
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5057 - loss: 0.4907
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5089 - loss: 0.4893
[32m][0m][37m-----][0m [1m1s][0m 262ms/step - accuracy: 0.5108 - loss: 0.4885
[32m][0m][37m-----][0m [1m0s][0m 262ms/step - accuracy: 0.5121 - loss: 0.4882
[32m][0m][37m-----][0m [1m0s][0m 262ms/step - accuracy: 0.5135 - loss: 0.4877
[32m][0m][37m-----][0m [1m0s][0m 262ms/step - accuracy: 0.5145 - loss: 0.4872
[32m][0m][37m][0m [1m0s][0m 785ms/step - accuracy: 0.5153 - loss: 0.4868
[32m][0m][37m][0m [1m71s][0m 2s/step - accuracy: 0.5159 - loss: 0.4864 - val_accuracy: 0.5771 - val_loss: 0.4432
Epoch 2/15
[1m 1/17][0m [32m][0m][37m-----][0m [1m5s][0m 348ms/step - accuracy: 0.6250 - loss: 0.5135
[32m][0m][37m-----][0m [1m3s][0m 263ms/step - accuracy: 0.6406 - loss: 0.5000
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.6493 - loss: 0.4905
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.6471 - loss: 0.4855
[32m][0m][37m-----][0m [1m3s][0m 262ms/step - accuracy: 0.6390 - loss: 0.4820
[32m][0m][37m-----][0m [1m2s][0m 263ms/step - accuracy: 0.6288 - loss: 0.4823
[32m][0m][37m-----][0m [1m2s][0m 263ms/step - accuracy: 0.6283 - loss: 0.4759
[32m][0m][37m-----][0m [1m2s][0m 262ms/step - accuracy: 0.6337 - loss: 0.4643
[32m][0m][37m-----][0m [1m2s][0m 263ms/step - accuracy: 0.6405 - loss: 0.4535
[32m][0m][37m-----][0m [1m1s][0m 263ms/step - accuracy: 0.6442 - loss: 0.4463
[32m][0m][37m-----][0m [1m1s][0m 263ms/step - accuracy: 0.6464 - loss: 0.4405
[32m][0m][37m-----][0m [1m1s][0m 263ms/step - accuracy: 0.6485 - loss: 0.4357
[32m][0m][37m-----][0m [1m1s][0m 263ms/step - accuracy: 0.6491 - loss: 0.4321
[32m][0m][37m-----][0m [1m0s][0m 263ms/step - accuracy: 0.6493 - loss: 0.4297
[32m][0m][37m-----][0m [1m0s][0m 263ms/step - accuracy: 0.6497 - loss: 0.4276
[32m][0m][37m-----][0m [1m0s][0m 263ms/step - accuracy: 0.6497 - loss: 0.4261
[32m][0m][37m][0m [1m0s][0m 254ms/step - accuracy: 0.6497 - loss: 0.4247
[32m][0m][37m][0m [1m7s][0m 410ms/step - accuracy: 0.6496 - loss: 0.4235 - val_accuracy: 0.6812 - val_loss: 0.3952
Epoch 3/15

Average Validation F1-Score across all folds:

0.09656506984658127

Average Validation MSE across all folds:

0.35908857444922127

Average Validation Accuracy across all folds:

0.7789444462458293

Best Hyperparameters Found:

Best Fold: 2

Best Trial Score: 0.893750011920929

Best Hyperparameters: {'conv_filters': 192, 'conv_activation': 'relu', 'units': 128, 'dense_activation': 'relu', 'dense_units': 128, 'dropout': 0.5, 'optimizer': 'adam', 'learning_rate': 0.001, 'batch_size': 32, 'epochs': 100, 'validation_split': 0.2, 'seed': 42}